Confounding

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In this article, I attempt to synthesize a coherent framework for the different sources of associations between genetic variants and phenotypes. In the literature, many of the concepts described here are referred to under different terms. I try to be consistent with my usage of terms in this article and also give alternate terms that are commonly used in the literature. For the most part, I am simpy synthesizing and/or paraphrasing existing literature, which I reference where applicable. A reference that helped me immensely in organizing my thinking for this entire document is [**?**]. I use the terms "trait" and "phenotype" interchangeably.

This is a rough draft and some of the language and sentence structure here is clunky. The overall structure of sections is probably suboptimal and subject to change. I acknowledge that parts of this article may get too much into the weeds of certain definitions/concepts.

1 Generative Model for a Phenotype

I begin by defining the broadest generative model for a trait Y that I can think of:

$$Y_i = f(G_i, E_i)$$

where for some individual i, G_i denotes all of their genetic information and E_i denotes all of their non-genetic (often referred to as environmental) information. Here, f denotes some function that takes in genetic and environmental information and outputs a trait value. In essence, f determines exactly how the trait comes about in the real world by translating its preceding physical processes into a mathematical formula.

Genetic information can be distilled down into distinct units of genetic variation, which I refer to as genetic **variants**. For a given genetic variant, an **allele** refers to a possible value of that variant. Different types of genetic variants exist, such as:

- **Single-nucleotide polymorphisms (SNPs)**: a single nucleotide letter change in a DNA sequence, such as having an A allele vs a T allele at some location in the genome. Another term for SNP is **single-nucleotide variant (SNV)**. Originally, SNP was supposed to refer to SNVs whose minor allele is "common" in a population, although such distinction is rarely applied in modern papers.
- **Insertions and deletions (indels)**: either the presence (insertion) or absence (deletion) of a string of nucleotides at some location in the genome.

- **Structural Variants (SVs)**: [To-do]
- **Copy-Number Variants (CNVs)**: [To-do]
- [...]

Regardless of the type of genetic variants being considered, we can parameterize our generative model as being a function of M genetic variants and an environmental term:

$$Y_i = f(g_{i,1}, ..., g_{i,M}; E_i)$$

1.1 Causality

A genetic variant g_j has a **causal effect** on a phenotype if changing g_j while holding all other terms in f fixed at any values leads to a change in the phenotypic value Y. This definition allows for interactions between genetic variants and environments since a genetic variant can only be non-causal if for all possible values of the remaining genetic and environmental terms, changing g_j does not change Y.

1.1.1 Environmental mediation of a causal genetic effect

A genetic variant being causal does not mean its effect is not environmentally mediated. For example, suppose a genetic variant causes an individual to have ginger hair. In a society where those with ginger hair are discriminated against and prevented from getting an education, this genetic variant would also be causal for having lower educational attainment. However, it would be silly to interpret this causality as meaning that the educational-attainment-lowering causal effect of this genetic variant is an immutable biological property. Later, I propose a way to think about environmentally-mediated effects as an environmental interaction.

1.2 Form of f

To do meaningful work with such a generative model, we have to impose a form onto f, which is typically done by adopting two assumptions: **additivity** and **linearity**. We also simplify the expression for the environmental effect.

1.2.1 Additivity

The additivity assumption states that the effect of a specific genetic variant g_j is independent of all other genetic variants and environmental factors. In other words, it assumes that interactions between causal factors do not exist, such as gene x gene (GxG) or gene x environment (GxE) interactions. Therefore, predicting the change in phenotype caused by a change in a genetic variant only requires knowledge of that genetic variant. We can formulate this as each genetic variant having its own function f_j that determines its effect on the phenotype. All factors' effects are then added together to determine the phenotypic value, hence the term "additivity":

$$Y_i = f_1(g_{i,1}) + ... + f_M(g_{i,M}) + f_E(E_i)$$

1.2.2 Linearity

The linearity assumption states that the form of each variant's effect function f_j is linear. For example, if a SNP is encoded as the the number of alternate alleles (0, 1, or 2) that an individual possesses in their genome for a locus, i.e. an **allele count**, then relative to having 0 alternate alleles, having 2 alleles should have double the effect on the phenotype as having 1 alternate allele. This precludes the presence of dominance/recessive effects of a variant. Updating our generative model under this assumption looks like:

$$Y_i = \beta_1 g_{i,1} + ... + \beta_M g_{i,M} + f_E(E_i) = \sum_j^M \beta_j g_{i,j} + f_E(E_i)$$

where β_j denotes the linear causal effect of variant j on the phenotype.

1.2.3 Noise and Environmental Component

Phenotypic generative models often consider the effect of **noise**, ϵ, which I define as a causal factor on a phenotype that has no antecedent that causes it. In other words, it is a causal factor whose value is generated at random for an individual such that it cannot be predicted beforehand. Without getting too philosophical, I believe everything is caused by something else in a deterministic fashion, and therefore there is no such thing as true noise. However, in practice, there are tons of environmental causal factors that conceivably cannot be measured and thus could be realistically modelled as noise. I don't need to make assumptions about the distribution of this noise component ϵ. However, due to noise typically capturing the cumulative contribution of many small immeasurable causal factors, it is typically assumed to be normally distributed.

For this article, I use the term $e_i = f_E(E_i)$ to denote the **environmental component** of a trait, which may be non-independent of other (genetic) factors. In contrast, ϵ_i is the **noise component** and aggregates the effects of environmental factors that are (in practice) independent of all other factors. At this point, we are not making assumptions about the environmental component being independent of genetics, therefore we are not dealing with noise. Our final generative model assumes additivity and linearity of genetic effects:

$$Y_i = \sum_j^M \beta_j g_{i,j} + e_i$$

Furthermore, the aggregate of additive genetic effects, $A_i = \sum_j^M \beta_j g_{i,j}$, is the **additive genetic component**, or just genetic component, of the trait.

1.2.4 Aside: Environmentally-mediated genetic causal effects

Earlier, I presented an example of a genetic variant that causes ginger hair that is also causal for educational attainment in the circumstance where gingers are discriminated against in society. It may seem unusual to refer to this variant as being causal for educational attainment since we could easily conceive of a non-discriminatory society where this would not be the case, and intuitively, such a society is the "default". I square this by going back to the most general generative model prior to any additivity ssumptions and arguing that there is an interaction effect between the genetic variant and the environmental state of being in a discriminatory society. That is:

$$f(g|E = \text{not discriminatory}) = 0 \quad \text{and} \quad f(g|E = \text{discriminatory}) \neq 0$$

Upon thinking about it in this context, I am troubled by the realization that all **main** (i.e. non-interacting) effects of genetic variants could be theoretically thought of as interactions. For example, a genetic variant that simply accelerates bone growth is, in an abstract sense, equally interacting with the environment, where the environment could be something like the presence of bone-forming molecular mechanisms that are needed for bone growth to exist in the first place. Without such mechanisms, the genetic variant would not accelerate bone growth. In reality, all surviving humans need and therefore already possess those bone-forming molecular mechanisms, so it may appear obtuse to to describe this as a gene-environment interaction. Furthermore, by allowing any theoretically possible environment to be considered, any genetic variant could conceivably be defined as causal for any phenotype.

In my opinion, the sound way of squaring these perspectives is to require the generative model to be defined for a particular population. If the environmental factor that could theoretically interact with a genetic variant does not vary in the population, then it is fair to think about the genetic variant's effect as being a main effect. Under this definition, the ginger-causing genetic variant would have a causal main effect on educational attainment in a discriminatory society and would not be causal at all in a non-discriminatory society. As long as we define what the population of our generative model is, there is no issue here. If our population is a mix of both societies, then there is a gene-environment interaction effect, since the environmental factor of being in a discriminatory-society does vary within the population. This also makes it easier to apply the additivity assumption, since we just have to think, "Do the environmental factors that could potentially interact with this genetic variant meaningfully vary within the population?"

2 Overview on Confounding

The causal effect of a genetic variant j is denoted by β_j, which we are interested in estimating accurately. As stated, we assume this causal effect is both linear and independent of other effects (such that it is additive). The naive approach to estimating β_j is to obtain a sample of individuals from the population and run a regression of the phenotype on the genetic variant:

$$Y_i \sim \hat{\beta}_j g_{i,j}$$

If we think of genotypes as being random variables, then according to ordinary linear regression, the expected value of the estimated effect is:

$$\mathrm{E}[\hat{\beta}_j] = \frac{\mathrm{Cov}[g_j, Y]}{\mathrm{Var}[g_j]}$$

If this expected value equals the true causal effect, $\mathrm{E}[\hat{\beta}_j] = \beta_j$, then our estimate is not confounded (i.e. it is unbiased). On the contrary, if this expected value does not equal the true causal effect, $\mathrm{E}[\hat{\beta}_j] \neq \beta_j$, then our estimate is confounded (i.e. it is biased).

2.1 Producing non-confounded estimates

We can expand the above formula for $\mathrm{E}[\beta_j]$ and determine what assumptions we need to make in order to obtain an unbiased estimate of β_j:

$$
\begin{aligned}
\mathrm{E}[\hat{\beta}_j] &= \frac{\mathrm{Cov}[g_j, Y]}{\mathrm{Var}[g_j]} \\
&= \frac{\mathrm{Cov}[g_j, \sum_k^M \beta_k g_k + e]}{\mathrm{Var}[g_j]} \\
&= \frac{(\sum_k^M \mathrm{Cov}[g_j, \beta_k g_k]) + \mathrm{Cov}[g_j, e]}{\mathrm{Var}[g_j]}
\end{aligned}
$$

If we assume that genetic variant j is uncorrelated with all other genetic variants as well as the environmental component, then all covariance terms become zero except for the covariance between the genetic variant and its own genetic effect:

$$
\begin{aligned}
&= \frac{\mathrm{Cov}[g_j, \beta_j g_j]}{\mathrm{Var}[g_j]} \\
&= \frac{\beta_j \mathrm{Var}[g_j]}{\mathrm{Var}[g_j]} \\
&= \beta_j
\end{aligned}
$$

These assumptions of independence between genetic variants and with the environment are the easiest way to achieve an unbiased estimate of β_j. However, it is theoretically possible for the genetic variant to be correlated with other factors in such a way that their biasing effects cancel out and leave the final estimate free of confounding. In practice, this is unlikely. This exercise reveals that there are two types of confounding: genetic and environmental.

3 Genetic Confounding

Genetic confounding is the non-independence between the focal genetic variant and other genetic variants: $\mathrm{Cov}[g_j, g_k] \neq 0$. [Does the correlated genetic variant have to be causal itself to induce genetic confounding?] There are many ways that genetic variants can become correlated in a population.

Before proceeding further, it is important to distinguish when two genetic variants are in cis vs in trans. Depending on the context, a genetic variant can refer to the unit of genetic variation that exists on a single chromosome or to the aggregate of those units across the two copies of each chromosome that diploids carry. In the former context, two genetic variants that lie on the same chromosome are said to be **in cis**. In contrast, two genetic variants that lie on different chromosomes are said to be **in trans**. This includes the two copies of the same unit of genetic variation as well as two genetic variants that are in different chromosome numbers altogether (e.g. chromosome 1 vs 22). The distinction is important because two variants that are in cis are inherited from the same parent (although they do not have to also be in cis in the parent's genome), while variants that are in trans on the same chromosome number are inherited from different parents. Relationships between variants in cis reflects the structure of the genomes in the population, while relationships between variants in trans further reflect the patterns of mating in the population. Because recombination causes variants on the same chromosome number that are in trans to be in cis in the subsequent generation, the structure of the genomes in the population reflects the prior patterns of mating in the population.

3.1 Linkage

Linkage refers to the tendency of two genetic variants that are in cis to segregate into the same gamete during meiosis. Typically, this is a function of physical proximity in the genome, since contiguous segments of a chromosome are segregated together during meiosis until a crossover event occurs, flipping which chromosome forms the gamete. This leads to two alleles that are found in cis on some chromosome to continue to be present in cis in subsequent generations, causing a correlation between them. For example, suppose there are two genetic variants (both biallelic: A/T) physically close to each other on a chromosome, and in a population, two haplotypes are observed: AA and TT. The two genetic variants are correlated with each other because the A allele for the first variant is only ever found alongside the A allele for the second variant, and vice-versa for the T alleles. Because the two variants are in strong linkage, it is unlikely for a crossover event to occur between the two variants, which would otherwise cause a heterozygote to produce an AT or TA haplotype that could be inherited by their offspring and thus lower the correlation between the two variants. The **recombination fraction** between two variants, c_{jk}, quantifies the probability of an odd number of crossover events occurring between the two loci, which causes two alleles that are in cis to segregate to different gametes. A recombination fraction of $c_{jk} = 0.5$ implies non-linkage, since it is equally likely for two alleles in cis to segregate together vs apart.

Theoretically, all genetic variants on the same chromosome are in non-linkage with each other, since it is possible for the entire chromosome to segregate together without a crossover event occurring. In contrast, genetic variants on different chromosomes are never in linkage with each other because parent-of-origin information is lost in meiosis such that chromosomes that were inherited from the same parent are not more likely to segregate together. In practice, the field assumes a fuzzy recombination fraction threshold for which two variants below the threshold are considered sufficiently linked and are said to be **tagging** or **local** to each other. Typically, the local region of a focal variant does not exceed tens to hundreds of kilobases.[**?**]

Linkage is not a population-level process, and whether it actually constitutes as confounding is undecided within the field. For example, suppose two genetic variants are directly adjacent to each other in the genome such that there is never a crossover event between, and thus they are perfectly correlated with each other. However, only the first variant is causal for some trait. In a population-level association analysis, distinguishing which variant is causal is impossible, since the causal effect of the first variant will be perfectly tagged by the second variant. Even within siblings of a parent that is heterozygous at these variants, the effect of each variant cannot be separated since they are always inherited together even within families. Should this trouble us? If our goal is solely prediction or quantifying the aggregate effect of genetic variants, then no, either variant can give us the information we need. But if our goal is to understand biology, then the distinction matters, since molecular mechanisms must be interact with each variant in different ways.

3.1.1 Direct genetic effects

Direct genetic effects (DGEs) refer to the aggregate causal effect of all variants tagged by a focal variant, including itself. The field seems to concretely agree on this definition and use it consistently, but note that a variant have a DGE does not require it to have a causal effect. Furthermore, the aggregate of causal effects is not the same as the aggregate of DGEs, since the latter includes double-counting of the same tagged causal variants. A way to unify these perspectives is to think of individual variants as tagging a **locus** (a region of tightly linked variants) and assigning DGEs to loci, not the variants themselves. However, for this article, I try to maintain the original definition of DGEs that applies to variants and not loci.

To-do: write the expected $\hat{\beta}$ given the recombination fraction, or at least the expected covariance between genetic variants given the recombination fraction

3.2 Population Structure

Population structure is a form of non-random mating whereby preferential mating occurs based on subpopulation membership. For example, suppose a population actually consists of two subpopulations where individuals only mate with others belonging to their subpopulation. **Genetic drift**, the random fluctuations in allele frequencies over generations, will occur independently between each subpopulation, leading to distinct allele frequencies. In an abstract sense, each individual has their own set of allele frequencies from which their genotype was drawn from; population structure is present when these individual-level allele frequencies differ across the population. If two alleles rise in frequency in one subpopulation while falling in frequency in the other subpopulation, then in the aggregate population, the alleles become correlated with each other because they signal subpopulation membership.

The consequence of population structure is that a proband's parents are going to have more similar genotypes than expected under random mating. However, unlike assortative mating (discussed next), the expected similarity is largely the same across the entire genome and therefore not specific to any particular trait (at least in the absence of population stratification, discussed later). Technically, some regions of the genome can be under accelerated genetic drift due to background selection and this could theoretically correlate with a trait's genetic architecture. However, the process driving this similarity remains subpopulation membership.

By random chance, the genetic component of a trait may correlate with subpopulation membership because the trait-increasing causal alleles tended to increase more in one subpopulation vs the other. This means if the focal trait is differentiated across subpopulations, it will be correlated with other variants that are differentated across subpopulations, whose causal effects with a trait will be picked up by the focal variant's association with the trait. Because (neutral) genetic drift is symmetric with regards to effect size, population structure should not systematically bias effect estimates towards or away from zero on expectation, but for some given population structure, it can (by chance).

To-do: Mathematically write out how population structure biases effects

3.3 Assortative Mating

Assortative mating (AM) is a form of non-random mating whereby preferential mating occurs based on the two potential mates' trait values. **Direct assortatment** refers to when the focal trait of interest is the trait that mating is sorting on (the **sorting factor**), while **indirect assortatment (iAM)** refers to when the focal trait of interest is simply correlated with the sorting factor [**?**]. **Positive assortative mating** refers to when individuals preferentially mate with individuals with similar trait values, while **negative assortative mating** refers to when individuals preferentially mate with individuals with dissimilar trait values. The former is much more common in human populations and thus is often used interchangeably with the term assortative mating.

The strength of assortative mating is typically quantified by the correlation between mates' phenotypic values, $r = \text{Corr}[Y_{m(i)}, Y_{p(i)}]$, where $m(i)$ and $p(i)$ refer to the mother and father of proband i, respectively. This correlation between trait values also induces a correlation between any factors that are correlated with the trait values, such as genetic variants with a causal effect on the trait, in the same direction as the assortative mating. So, the trait-increasing allele of a causal variant will be positively correlated with the trait-increasing allele of another causal variant under positive assortative mating. Consequently, an association between a causal genetic variant and a trait under assortative mating will not only pick up its own causal effect, but also the causal effects of all other variants for that trait, biasing estimates away from zero.

To-do: Mathematically write out how correlation between trait increasing alleles arises from AM. See [Yengo et al. 2018 NatGen]

Cross-trait assortative mating (xAM) refers to correlations between two traits across mates (e.g. father's height and mother's education). This can easily occur when the two traits share causes. It is unclear to me whether the consequences of xAM on variants' correlations with each other depends on just the shared causal structure, or additionally on there being a cross-trait mating preference to begin with.

In an abstract sense, population structure is an example of assortative mating, where the trait being sorted on is subpopulation membership and variants' causal effects on the trait is the differentiation of allele frequencies between subpopulations. Like population structure, correlations between genetic variants start off as being entirely in trans due to non-random mating, but become in cis correlations over time as recombination places variants from different parents onto the same chromosome in subsequent generations.

3.4 Selection / Ascertainment / Migration

Here, I describe very different processes that act similarly in the context of genetic confounding.

3.4.1 Stabilizing and Disruptive Selection

Selection refers to when the number of offspring produced by an individual depends on their trait values. Some forms of selection can induce correlations between causal variants, such as **stabilizing selection** (deviations from some optimal trait value are selected against) and **disruptive selection** (deviations from some sub-optimal trait value are selected for).

In stabilizing selectioin, by selecting against extreme values of a trait, the co-occurrence of variants with effects in the same direction is less frequent in the mating population. As a consequence, trait-increasing alleles will become negatively correlated with each other among gametes. When sampling a population, this negative correlation will be present between variants in cis, and if selection acts on the probability of survival prior to sampling, it will also be present between variants in trans. [Low-key unsure about this]. When performing an association analysis on a trait that has undergone stabilizing selection, this should bias estimated effects towards zero.

Disruptive selection should have the opposite effect, inducing positive correlations between trait-increasing alleles and biasing estimates away from zero.

3.4.2 Ascertainment

Ascertainment refers to when the inclusion of an individual into a sample depends on their trait values. For example, it is known that participants in the UK Biobank tend to be wealthier and healthier than the average Brit [To-do: ref]. Therefore, there is ascertainment for wealth and health.

Ascertainment is akin to conditioning on a collider (inclusion in the sample) and thus can induce correlations between genetic variants, an example of Berkson's paradox. For example, if two genetic variants are causal for wealth, and only rich individuals can become part of our sample, then the two wealth-increasing alleles will appear negatively correlated within our sample.

Ascertainment can also be thought of as directional selection occurring on a single generation and only acting on individuals' survival. However, unlike selection, ascertainment does not affect subsequent generations. Ascertainment induces correlations in variants both in cis and in trans. It biases estimates towards zero due to the negative correlation of trait-increasing alleles.

3.4.3 Differential Directional Selection and Migration

In a structured population, we previously assumed that allele frequencies fluctuated symmetrically. However, additional processes can make this not the case.

If the subpopulations differ in the strength of directional selection for some trait (e.g. subpopulation 1 is being selected for height while subpopulation 2 isn't), then the differences in the average genetic component of each subpopulation will reflect these differences in selection. As a consequence, variants that signalled subpopulation membership will now also become correlated with the trait undergoing differential selection. For a non-causal variant, the expected bias on its estimate is still zero since non-causal variants' allele frequencies are only subject to genetic drift and not directional selection. However, for causal variants, the trait-increasing alleles will rise in frequency alongside other trait-increasing alleles due to the shared selection happening on them within select subpopulations. As a result, causal variants' population-level estimates will be biased away from zero.

Similarly, if migration between subpopulations depends on individuals' trait values, the same exact process occurs. For example, if migrants from subpopulation 2 to subpopulation 1 tend to be wealthier, then it should drive up the average genetic component of subpopulation 1. Wealth-increasing alleles will become positively correlated with each other because they each signal membership into subpopulation 1 where potential mates also tend to have these wealth-increasing alleles.

Directional selection by itself in an unstructured population is not expected to cause correlations between genetic variants except beyond what was described in the ascertainment section.

4 Environmental Confounding

Environmental confounding is the non-independence between the focal genetic variant and the environmental component: $\text{Cov}[g_j, e] \neq 0$.

4.1 Population Stratification

4.2 Indirect Genetic Effects

4.2.1 Genetic Nurture

4.2.2 Sibling indirect genetic effects

4.2.3 Dynastic Effects